

More than 67.3195% of the zeros of the Riemann zeta function are simple and lie on the critical line

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Abstract

Let $N(T, 2T)$ count the zeros of the Riemann zeta function with ordinates in $(T, 2T]$, with multiplicity, and let $N_0^s(T, 2T)$ count the simple zeros on the critical line in that interval. We prove

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.673195198901 \dots > \frac{673195}{10^6}.$$

The proof combines the analytic framework of Claude [1] (Theorem D and its Lean artifact), the stability refinement of the rank–trace argument introduced in [3], and the re-optimized window, position-weighted seven-point inequality, and sharp block profile of [4]. The single new mathematical ingredient is a strengthened certified constant for the weighted seven-point inequality of [4]: we certify $F \geq 509/100000$ where $1/200$ was certified before, and re-optimize the block length. All finite claims are certified by Arb interval arithmetic in two independent runs; the deduction is reproduced in full.

1 Statement and imported inputs

Throughout, write $N := N(T, 2T)$ for the number of zeros of ζ with ordinates in $(T, 2T]$, counted with multiplicity, and $S := N_0^s(T, 2T)$ for the simple zeros on the critical line in that range.

Theorem 1.1.

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{250 H_{\text{cert}} - \eta \frac{3}{1150} \cdot 249}{250 - R} = 0.6731951989015205 \dots > \frac{673195}{10^6},$$

where $H_{\text{cert}} = 672457/10^6$, $A = 31049/25000$, $R = 2\sqrt{A} - 1$, and $\eta = R/A$.

The proof template is that of [4], which in turn builds on [3] and [1]. We import the following two statements exactly as they are established (and used) there; everything else in this paper is self-contained.

Imported Input 1.2 (Normalized ordinates and total span). *Put $\ell_T = \log(T/(2\pi))$ and, for a zero of ordinate γ , put*

$$x_\gamma = \frac{\ell_T(\gamma - T)}{2\pi}.$$

After the deletion of $o(N)$ exceptional zeros, the retained simple zeros have normalized ordinates $x_1 < \dots < x_{S^\circ}$ with $S^\circ = S - o(N)$ and

$$x_{S^\circ} - x_1 \leq \frac{T\ell_T}{2\pi} = N + o(N). \tag{1}$$

The final equality is the Riemann–von Mangoldt formula.

Imported Input 1.3 (Stability counting inequality). *Let $v : [-\frac{1}{2}, \frac{1}{2}] \rightarrow [\frac{3}{4}, 1]$ be an even trigonometric polynomial (an admissible window profile), used in the test family of [1, §2] with the standard boundary ramp. Write*

$$K_v(x) = \int_{-1/2}^{1/2} v(t) \cos(2\pi xt) dt, \quad k_v = \frac{K_v}{K_v(0)}, \quad H(v) = 2 - \frac{1}{c_1(v)},$$

where $c_1(v) = (\int v)^2 / (\int v^2 + \iint |s - s'| v(s) v(s') ds ds')$. Then

$$S \geq H(v) N + \text{tr } \Psi(M) - o(N), \quad (2)$$

where M is the Gram matrix of the retained simple-zero atoms, $\Psi(t) = (t - 1)^2$ for $0 \leq t \leq 2$ and $2t - 3$ for $t \geq 2$, and the entries of M satisfy $M_{ij} = k_v(x_i - x_j) + o(1)$ uniformly for normalized ordinate differences in compact sets, the exceptional $o(N)$ zeros being deleted.

We use one retained set for both imported inputs, enlarging the exceptional set to the union of the two $o(N)$ sets if necessary; it still has size $o(N)$.

Input 1.3 is the stability form of the rank–trace argument: the analytic content (explicit formula, Gabor trace asymptotics for admissible windows, tail estimates, Gram-entry asymptotics) is from [1] and its Lean artifact [2]; the retention of the defect $\text{tr } \Psi(M)$ is the contribution of [3], audited independently in [4] and formalized in Lean in [5]. The deduction below uses (2) only through the block/pinching mechanism of Section 5.

2 The window

We use the window of [4] unchanged:

$$v(s) = \sum_{j=0}^6 c_j \cos(\omega_j s), \quad \omega = (\sqrt{2}, 2\pi, 4\pi, 6\pi, 8\pi, 10\pi, 12\pi), \quad (3)$$

$$c = (10^9, 3322500, -7609135, 1190194, -731476, -1680572, 1141360)/10^9.$$

Proposition 2.1 (Certified window bounds). *On $[-\frac{1}{2}, \frac{1}{2}]$ one has $\frac{3}{4} \leq v \leq 1$ (certified enclosures: $\min v = 0.750213\dots$, $\max v = 0.995632\dots$), and*

$$H(v) = 0.67245704141454\dots \geq H_{\text{cert}} := \frac{672457}{10^6}.$$

Verification. Arb interval arithmetic over exact rational data; the enclosures above are reproduced on every run of the **fast** module of the verifier (see Section 6). This reproduces the corresponding certificate of [4]. $\square$

3 The sharp block profile

Lemma 3.1. *Let $G \succeq 0$ be Hermitian with $E := 2 \sum_{i < j} |G_{ij}|^2$. Then*

$$\text{tr } \Psi(G) \geq h(E), \quad h(E) = \begin{cases} E, & 0 \leq E \leq 1, \\ 2\sqrt{E} - 1, & E \geq 1, \end{cases}$$

and h is sharp.

Proof. For $\lambda \geq 0$ write $x = (\lambda - 1)^2$. Since $\Psi(\lambda) = (\lambda - 1)^2 - ((\lambda - 2)_+)^2$ and, for $\lambda \geq 1$, $(\lambda - 2)_+ = (\sqrt{x} - 1)_+$ while for $0 \leq \lambda < 1$ both $(\lambda - 2)_+ = 0$ and $x \leq 1$, one has the exact identity $\Psi(\lambda) = g((\lambda - 1)^2)$ with

$$g(x) = \begin{cases} x, & 0 \leq x \leq 1, \\ 2\sqrt{x} - 1, & x \geq 1. \end{cases}$$

The function g is nonnegative, increasing, concave, and $g(0) = 0$; hence g is subadditive, so for the eigenvalues λ_i of G ,

$$\text{tr } \Psi(G) = \sum_i g((\lambda_i - 1)^2) \geq g\left(\sum_i (\lambda_i - 1)^2\right) = g(\|G - I\|_F^2) \geq g(E),$$

using that $\|G - I\|_F^2 = \sum_i (\lambda_i - 1)^2$ and that the off-diagonal part alone contributes E . For sharpness, put $t_n = \sqrt{E(n-1)/n}$, $a_n = 1 + t_n$, and $b_n = 1 - t_n/(n-1)$, taking n large enough that $b_n \geq 0$. With $u_n = n^{-1/2}(1, \dots, 1)$, set $G_n = b_n I + (a_n - b_n)u_n u_n^*$. This matrix is positive semidefinite with constant diagonal one and off-diagonal energy exactly E . Its eigenvalues are $a_n, b_n, \dots, b_n$, and therefore

$$\text{tr } \Psi(G_n) = \Psi(1 + t_n) + (n-1)\Psi(1 - t_n/(n-1)) \longrightarrow h(E).$$

Thus no larger lower profile is possible even among correlation matrices. $\square$

Corollary 3.2 (Chord bound). *For $0 \leq E \leq A$ one has $h(E) \geq \frac{h(A)}{A} E$; and h is nondecreasing.* $\square$

4 The certified weighted seven-point inequality

Let $w = k_v^2$ for the window (3), let $g_1, \dots, g_6 \geq 0$ be consecutive normalized gaps with partial sums $y_j = g_1 + \dots + g_j$ ($y_0 = 0$), and define

$$F(g_1, \dots, g_6) = \frac{1}{2300} \sum_{i=1}^6 g_i + \sum_{0 \leq i < j \leq 6} a_{ij} w(y_j - y_i), \quad (4)$$

with the reflection-symmetric exact rational weights a_{ij} of Table 1. Every span capacity is exactly $\sum_i a_{i,i+r} = 2$ ($r = 1, \dots, 6$).

$10^6 a_{ij}$	$j = 1$	2	3	4	5	6
$i = 0$	239252	528172	965879	1000000	1000000	2000000
1		381335	465776	34121	0	1000000
2			379413	12104	34121	1000000
3				379413	465776	965879
4					381335	528172
5						239252

Table 1: The pair weights of [4], used unchanged.

Proposition 4.1 (Certified local inequality). *For all $g_1, \dots, g_6 \geq 0$,*

$$F(g_1, \dots, g_6) \geq \varepsilon_0 := \frac{509}{100000} = 0.00509. \quad (5)$$

Verification. Exhaustive interval subdivision with Arb enclosures of w on a grid of mesh $1/4000$, exactly the decision procedure of [4] with the target raised from $1/200$ to $509/100000$: one-body pruning, per-box range-minimum lower bounds over all 21 pair separations, a certified convex tangent bound, and bisection of the widest coordinate; the run fails loudly at any unresolved terminal cell. The recorded run has 1,739,356 nodes and maximum depth 57; the weaker targets $127/25000$ and $1/200$ were additionally certified in independent runs. See Section 6 for the recorded certificates. (The observed minimum of F is ≈ 0.005091029 , at a lattice-like configuration.) $\square$

5 Deduction of the main theorem

Set

$$q = 6, \quad m = 250, \quad p = \frac{1}{2300}, \quad B_p = 6p = \frac{3}{1150},$$

$$A = \varepsilon_0(m - q) = \frac{509}{100000} \cdot 244 = \frac{31049}{25000} = 1.24196, \quad R = h(A) = 2\sqrt{A} - 1, \quad \eta = \frac{R}{A}.$$

Numerically $R = 1.22886518210501\dots$, $\eta = 0.98945632879079\dots$

5.1 Windows to blocks

Let B be a block of m consecutive retained simple zeros with normalized ordinates $y_1 < \dots < y_m$, Gram matrix G_B , and $E_B = 2 \sum_{i < j} |(G_B)_{ij}|^2$. Summing (5) over the $m - q$ consecutive seven-point windows of B : a pair spanning r gaps lies in at most $7 - r$ windows and its weight totals at most the span capacity 2; each gap lies in at most 6 windows, contributing at most $6p \sum g_i = B_p \text{span}(B)$. By the nonnegativity of the weights, this gives the purely kernel-level bound

$$2 \sum_{i < j} w(y_j - y_i) + B_p \text{span}(B) \geq A. \quad (6)$$

5.2 Blocks to defect

We claim

$$\text{tr } \Psi(G_B) + \eta B_p \text{span}(B) \geq R - o(1). \quad (7)$$

Put

$$L := A/B_p = 714127/1500.$$

If $\text{span}(B) \geq L$, (7) is immediate: $\text{tr } \Psi(G_B) \geq 0$ and $\eta B_p \text{span}(B) \geq \eta A = R$. We may therefore suppose that $\text{span}(B) < L$. Every pair separation in B then lies in the fixed compact interval $[0, L]$. The Gram-entry asymptotic of Input 1.3 is uniform there, and hence

$$2 \sum_{i < j} w(y_j - y_i) \leq E_B + o(1).$$

Together with (6), this gives

$$E_B + B_p \text{span}(B) \geq A - o(1). \quad (8)$$

If $E_B \geq A$, then Lemma 3.1 and monotonicity give $\text{tr } \Psi(G_B) \geq h(E_B) \geq h(A) = R$. If $E_B < A$, the chord bound (Corollary 3.2) gives $\text{tr } \Psi(G_B) \geq \eta E_B$, and (8) yields $\eta E_B + \eta B_p \text{span}(B) \geq \eta A - o(1) = R - o(1)$. Because m and L are fixed, the $o(1)$ here is uniform over the small-span blocks; the large-span case has no asymptotic error.

5.3 Pinching and averaging

Exactly as in [3, §5] (and [4, §4]): for each of the m offsets, partition the retained simple zeros into consecutive blocks of size m . Pinching does not increase $\text{tr } \Psi$ because $X \mapsto \text{tr } \Psi(X)$ is convex and unitarily invariant; hence $\text{tr } \Psi(M)$ dominates the sum of the block defects. Since there are $O(N)$ blocks, the uniform per-block $o(1)$ above sums to $o(N)$. Summing (7) over the full blocks of one offset and averaging over the m offsets therefore gives, with the retained ordinates of Input 1.2, each interior gap lying in a block span for at most $m - 1$ of the m offsets,

$$\text{tr } \Psi(M) \geq \frac{R}{m} S - \eta B_p \frac{m-1}{m} N - o(N). \quad (9)$$

5.4 Conclusion

Insert (9) and Proposition 2.1 into (2):

$$S \geq H_{\text{cert}} N + \frac{R}{m} S - \eta B_p \frac{m-1}{m} N - o(N),$$

hence

$$\liminf_{T \rightarrow \infty} \frac{S}{N} \geq \frac{m H_{\text{cert}} - \eta B_p (m-1)}{m - R} = \frac{250 \cdot \frac{672457}{10^6} - \eta \cdot \frac{3}{1150} \cdot 249}{250 - R} = 0.6731951989015205755 \dots$$

The final arithmetic (including the enclosure of R and η and the comparison with $673195/10^6$) is itself certified in Arb by the verifier's **fast** module. This proves Theorem 1.1. $\square$

5.5 Sanity gates

With $(\varepsilon_0, p, m) = (1/200, 1/2300, 257)$ the same formula gives $0.673137630699 \dots$, the bound of [4]; with the unit-cap profile ($\eta = 1$, $R = A \leq 1$) and the parameters of [3] ($\varepsilon_0 = 19/5000$, $p = 1/3000$, $m = 269$, Montgomery–Taylor window) it gives $0.673008527927 \dots$, and with $\varepsilon_0 \rightarrow 0$ it degenerates to Theorem D of [1] ($0.672500703679 \dots$). The verifier additionally reproduces the certificates of both predecessors end-to-end.

6 Reproducibility

All finite claims (Propositions 2.1 and 4.1 and the final arithmetic) are certified by Arb interval arithmetic over exact rational inputs. Source, recorded certificates, and instructions are at

<https://github.com/npip99/zeta-zeros>

(the verifier is that of [4], vendored under MIT with only the design constants changed). Proposition 4.1 was certified in two independent runs of the identical decision procedure: the headline target $509/100000$ at grid $1/4000$ (1,739,356 nodes, depth 57) and again at grid $1/8000$ (1,736,620 nodes, depth 57), the intermediate target $127/25000$ certified twice (a run recorded here, 1,544,746-node independent run during the design exchange), and the original $1/200$ certificate of [4] reproduced end-to-end. The Lean 4 formalization of the framework ([2, 5]) covers Input 1.3's stability layer, the aggregation scheme, and the kernel-limit asymptotics; certifying the present constants in Lean is mechanical given the β -parametric capstone theorem of [5] and is left to a future artifact revision.

Credit

This note is a minimal increment on [4] (whose window, weights, profile, and verifier are used unchanged), which in turn extends [3] over the framework of [1]. The improvement was found and cross-verified in a Claude–GPT-5.6 collaboration on August 11, 2026.

References

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